

PROBLEMS CLASS 3

1. SIR model revisited

In problems class 1, you might have got far enough to see the popular SIR disease model:

$$\frac{dS}{dt} = -\beta IS, \quad (1)$$

$$\frac{dI}{dt} = \beta IS - \gamma I, \quad (2)$$

$$\frac{dR}{dt} = \gamma I, \quad (3)$$

$$N = S + I + R. \quad (4)$$

We're going to draw a phase portrait of (S, I) . But first, let's remember what this all means.

- (a) What do all these terms represent? Which term is reduced by social distancing? In this model, R is actually composed of two classes of patients: what are they? What is the assumption here about reinfection, and for which diseases is this correct?
- (b) Last time, we introduced the famous reproductive ratio R_0 . What was the intuition behind R_0 ? And what was the definition of R_0 in terms of the parameters we've seen here?
- (c) Consider only the equations for dS/dt and dI/dt and draw a phase portrait for (S, I) .
- (d) Mark the total population, N , onto the S axis. What can you say about the existence of one of your dI/dt nullclines to the left of $S = N$ in relation to the reproductive ratio R_0 ?
- (e) The main reason given for the lockdowns we experienced in 2020–21 was the potential pressure on the NHS, specifically the number of patients requiring hospitalisation at any given time.
 - (i) In our model, what could represent the number of patients requiring hospitalisation at any one time?
 - (ii) At what value of S is this number a maximum?
 - (iii) Divide two of our governing equations to find an expression for dI/dS . Assume that at the beginning of time, I_0 people are infected and the rest of the population is susceptible. Integrate this equation and show that if we assume $I_0 \ll N$, the maximum number of infected patients is given by

$$I_{\max} = N \left(1 - \frac{1 + \log R_0}{R_0} \right). \quad (5)$$

2. SIR with temporary immunity

So far, we have assumed that immunity is permanent (how has this been encoded into our model?) What if immunity is only temporary? We know that Covid-19 reinfections are possible.

- (a) Introduce a new term to our equations which represents the ‘movement’ of the population from the immune section, R , to the susceptible section, S .
- (b) Show that there are two equilibria of this new model, with one of them given by

$$S = \frac{\gamma}{\beta}, \quad I = \frac{N - \frac{\gamma}{\beta}}{1 + \frac{\gamma}{\delta}}, \quad R = \frac{N - \frac{\gamma}{\beta}}{1 + \frac{\delta}{\gamma}}, \quad (6)$$

where δ is the factor on your R terms in the governing equations. Is this different to our original SIR model, and if so, how?

- (c) By using the equation for N to eliminate R from your expression for dS/dt , write down two equations governing this system.
- (d) By writing down the Jacobian of this reduced system where appropriate, perform a linear stability analysis on the equilibria you have found, and state under what conditions each equilibrium is asymptotically stable/unstable/degenerate. (Hint: you may find the rules about determinants and traces useful.)
- (e) How might you explain this to the general public? (Assuming you are able to crop your slides to the correct aspect ratio.) [This last comment would have been considered humorous and topical in 2020; maybe you remember this...?]

3. Advection and diffusion

Last week we introduced the advection–diffusion equation for a concentration or population $c(\mathbf{x}, t)$,

$$\frac{\partial c}{\partial t} = -\nabla \cdot (c\mathbf{v}) + D\nabla^2 c + f. \quad (7)$$

- (a) In words, what is diffusion?
- (b) In words, what is advection?
- (c) What does f represent?
- (d) What does this equation look like in one spatial dimension, $\mathbf{x} = x$?